

The Formal Foundation: Concepts from Paper 1

Part 3 builds directly on the formal framework established in Part 1. For clarity, we summarize the core definitions and theorems here, utilizing the notation defined in the formal manuscript.

1. The Adversary Hierarchy (Ω)

Paper 1 formalized the “Scope of Threat” through a hierarchical taxonomy. This hierarchy allows precise definition of defensive scope.

- ▶ Ω_1 (**External**): The adversary controls inputs (e.g., prompt injection). The agent’s internal state is intact.
- ▶ Ω_2 (**Peripheral**): The adversary controls tools or RAG data (e.g., poisoned retrieval). The agent’s perception is compromised.
- ▶ Ω_3 (**Agent**): The adversary controls the agent’s weights or context (e.g., identity implementation). The agent itself is untrusted.
- ▶ Ω_4 (**Coordination**): The adversary controls a subset of the swarm (e.g., Sybil agents). The consensus mechanism is under attack.